Utkarsh Pandey

Generative AI Engineer | LLM Systems, RAG & Multi-Agent Architectures
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Education

Dr. D. Y. Patil Institute of Technology, Pimpri — Pune, Maharashtra
Bachelor of Engineering in Electrical Engineering — *Sept 2024 – July 2028*
- Expected Graduation: July 2028 (Currently 3rd Year)

Technical Skills

Languages: Python, JavaScript, SQL
AI/ML: LLMs, RAG, Multi-Agent Systems & Orchestration, LLM Evaluation, Vector Databases, Model Context Protocol (MCP), Prompt Engineering & Optimization, OpenAI API, GPT-4o, CLIP, Tavily API
Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy
Web Dev: Node.js, REST APIs, Supabase, React, Streamlit
Databases: SQLite, ChromaDB, FAISS, MariaDB Vector
Cloud & Deployment: Vercel, Render, Firebase
Developer Tools: Git, VS Code, Jupyter, Postman
Other: Data Structures & Algorithms, AI Automation

Experience

Co-Founder & AI Systems Lead — Jan 2026 – Present
Autonix – AI Automation Agency — *Pune, India (Remote)*
- Co-founded and own end-to-end AI systems architecture for an automation agency, delivering production AI systems to dental, aesthetic, and home-service clients across UK and UAE markets.
- Architected and deployed a production AI voice receptionist (Vapi, n8n, Google Calendar) with automated call-routing and booking orchestration, enabling 24/7 inbound coverage and eliminating manual scheduling for the client.

Full-Stack Developer Intern — Jan 2026 – Present
Own The Story (NGO) — *Remote*
- Own development of frontend UI and backend REST APIs for production web applications supporting the NGO's community outreach programs, collaborating cross-functionally through code reviews to ship production-ready features.

Projects

Forge – AI Support Investigation Platform | *Python, FastAPI, Next.js, ChromaDB, SQLite* | GitHub — 2026
- Built a full-stack AI platform (FastAPI backend, Next.js dashboard) so support and engineering teams can investigate tickets and GitHub Issues via natural language, using a deterministic planner that orchestrates allowlisted SQL analytics and local ChromaDB semantic retrieval over a locally-served BGE embedding model and an OpenAI reasoning layer.
- Engineered an LLM evaluation and observability layer – confidence scoring, retrieval-strategy transparency, and refusal on unsupported claims – validated through a 50-case Recall@5/hallucination-rate harness and 46-test suite to keep production outputs evidence-grounded.

Agentic Research PRO | *Python, LLMs, ChromaDB, Tavily API, FAISS* | GitHub — 2025
- Architected an autonomous multi-agent orchestration pipeline (7 specialized agents) automating end-to-end research: web search (Tavily), scraping, semantic chunking, embedding, vector storage, and structured report generation.
- Implemented FAISS-based vector search with semantic chunking and automated PDF export, including source citations and clickable URLs.

SkyVision – Multimodal Travel Search | *Python, CLIP, MariaDB Vector, Streamlit* | GitHub — 2025
- **MariaDB Hackathon Shortlisted:** Built multimodal search engine for airports and airline logos using CLIP embeddings and MariaDB Vector, enabling natural language and image-based visual similarity search.
- Deployed a production-style Streamlit inference interface achieving 85% relevance accuracy on 512-dimensional CLIP embeddings via cosine similarity.

Achievements & Certifications

Flipkart GRID Hackathon – Semifinalist: Advanced to the semifinal round among 35k+ participants.
VIBE HACKS 2025 – 2nd Place Winner: Secured 2nd place among 600+ teams with an innovative AI solution.
6th Rank – InterCollege Hackathon: Ranked 6th among 200+ teams in a competitive innovation challenge.